

Pre-Construction Feasibility & Site Planning

Geotechnical Soil Borings (Min. 2 per Footprint)

Establishes exact soil bearing strength (PSF/kPa) and high-water table depth to prevent differential settlement and structural cracking.

Topographic Grading & Cut/Fill Balance

Sets Finished Floor Elevation (FFE) relative to road crowns and high-water marks. Balances site earthworks to eliminate imported aggregate costs.

Regulatory Approvals Framework

Distinguishes Outline Approval (macro density & zoning) from Detailed Approval (structural framing, MEP, and fire safety execution).

Financial Risk Guardrail

Locks in a mandatory non-negotiable 15% hard-cost contingency buffer for unforeseen subsurface anomalies and material shifts.



Site Exploration & Soil Boring Rig

[IMAGE PLACEHOLDER: Soil sampling rig executing geotechnical borings on raw site terrain]

Substructure & Foundation Integrity

Subgrade Stripping & Proof-Rolling

Excavates down to virgin unyielding subgrade. Proof-rolls site with heavy equipment to catch soft pockets before formwork setup.

Concrete Rebar Clearance (2-Inch Rule)

Elevates steel rebar grids on concrete chairs to guarantee 2 inches of continuous concrete encasement, preventing moisture corrosion.

Hydration Curing & Compression Testing

Standard concrete requires 28 days of hydration to reach design compressive strength (PSI). Conducts slump tests on every truck load.

Tie-Beam & Hurricane Strap Integration

Pours continuous reinforced masonry tie-beams to lock walls into a rigid box. Embeds heavy hurricane straps directly into concrete tie-beams.



Rebar Grid & Concrete Chairs

[IMAGE PLACEHOLDER: Close-up detail of rebar matrix elevated on chairs inside foundation forms]

Superstructure Framing & Building Envelope

Load-Bearing Wall Framing & Headers

Aligns vertical studs directly over floor joists (advanced framing). Sizes structural headers over window and door openings to transfer roof loads.

Roof Truss Anchoring & Uplift Resistance

Fastens engineered roof trusses to top plates using heavy-duty hurricane ties. Ensures continuous load paths down to foundation tie-beams.

Weather-Resistive Barrier (WRB) Lap Rules

Installs housewrap horizontally, shingle-lapping upper layers over lower layers by minimum 6 inches to shed water outward.

Window Sill & Pan Flashing Integration

Applies self-adhered flexible sill pan flashing before window installation. Seals jamb flanges while leaving bottom sill open to drain.



Framing & Weather Sheathing

[IMAGE PLACEHOLDER: Structural wall framing with engineered trusses, housewrap, and window flashing detail]

MEP Infrastructure & Rough-In Mechanics

Drain-Waste-Vent (DWV) Slope Rules

Maintains continuous 1/4 inch per foot pitch for lines under 3 inches. Roof vent stacks balance pressure to prevent trap siphonage.

HVAC Manual J Load Calculations

Mandates ACCA Manual J sizing. Prevents oversized compressors that short-cycle, cooling quickly without removing humidity, causing mold.

Electrical Sub-Panel Phase Balancing

Equalizes current draw across both hot legs in single-phase panels to prevent neutral leg overload, voltage drops, and breaker trips.

Nail-Guard Protection & Pressure Testing

Installs steel plates where lines pass within 1-1/4" of stud edges. Hydrostatic tests supply lines at 100 PSI for 24 hours.



Rough-In Framing & MEP Routing

[IMAGE PLACEHOLDER: Commercial stud wall framing with rough plumbing, electrical conduit, and HVAC runs]

Enclosures & Drywall Mechanics

Acoustic Decoupling Channels

Mounts RC-1 resilient channels horizontally across studs to break physical contact and eliminate direct acoustic vibration transfer.

Subfloor Flatness & SLU Preparation

Limits subfloor variance to max 3/16" per 10-foot radius. Applies acrylic primer and foam expansion strips before pouring SLU.

Level 5 Drywall Skim Finish

Applies continuous joint compound skim coat across entire board surface. Mandatory under glossy paints or harsh side-lighting.

Continuous Vapor Barrier Placement

Positions continuous vapor retarder on the warm side of wall insulation to prevent moisture condensation within stud cavities.



Level 5 Drywall Application

[IMAGE PLACEHOLDER: Taped, seamless level 5 skim-coated wall assembly inside a building envelope]

Architectural Millwork & Custom Finishes

In-Wall Structural Wood Blocking

Installs continuous 2x8 solid lumber blocking between studs behind drywall for all wall-hung cabinetry and floating vanities.

45-Degree French Cleat Mounting

Secures heavy millwork with 45-degree interlocking bevel rails anchored directly into blocking, using gravity to pull cabinets tight.

MR MDF Cores & Balancing Sheets

Specifies Moisture-Resistant MDF in wet areas. Applies matching backer laminate to unexposed rear faces to prevent panel cupping.

Quartz Overhang Steel Supports

Limits unsupported 3/4" quartz overhangs to 8 inches. Overhangs exceeding 8" require 1/4" concealed structural steel flat bars.



Custom Cabinetry & Structural Blocking

[IMAGE PLACEHOLDER: Floating vanity millwork showing internal cleat mounting and quartz top installation]

Project Closeout, Inspections & Handover

Simultaneous Water Flow Testing

Runs all plumbing fixtures concurrently to verify real-world dynamic pressure drops and main stack drainage capacity under peak load.

Unconditional Lien Releases

Collects signed unconditional lien waivers from General Contractor, sub-trades, and material suppliers prior to final retainage payout.

Turnover Operations Binder

Compiles as-built architectural/MEP drawings, trade contact directories, equipment operation manuals, and warranty documentation.

Blue-Tape Punch List Clearance

Systematically marks and rectifies cosmetic wall/trim flaws and hardware defects prior to final Certificate of Occupancy sign-off.



Final Compliance & Turnover Audit

[IMAGE PLACEHOLDER: Site managers conducting blue-tape punch list walkthrough with compliance documentation]